Lesson notes · for you

Linear programming and the simplex method

Optimization · Level 4 · Lesson 1

Time	Materials
25 min	Your phone, or a computer with Colab; Graph paper and a ruler

Learning goals

- Write a decision as a linear program in standard form.
- Explain why the simplex method walks from corner to corner.
- Read binding constraints and shadow prices, and solve an LP with GLOP.

Activity: The garden on graph paper

1. Draw (8 min): On graph paper, x across and y up, 0 to 10. Draw $x + y = 8$, $2x + y = 14$ and $x + 3y = 18$, and shade the side that keeps all three.
2. Corners (6 min): Mark the corners (0, 0), (7, 0), (6, 2), (3, 5) and (0, 6) and work out $30x + 40y$ at each. The best is 290 at (3, 5).
3. Walk (4 min): Start at (0, 0). Move along an edge to a better neighbour each time; check that you can't improve once you reach (3, 5).
4. Prices (7 min): Redraw the hours line as $x + 3y = 19$ and find the new corner, (2.5, 5.5), worth 295: the shadow price is 5. Do the same with land ($x + y = 9$): 315, worth 25. Why is water worth nothing?

No graph paper? Use squared exercise-book paper, one square per bed. Check your corners against the lab "The best plan is at a corner".

Discussion

- Why does convexity let simplex stop without checking every corner?
- What does a zero shadow price tell a manager?
- When would you choose an interior-point method over simplex?

Check yourself

- Explain in two sentences why an extra can of water adds nothing to the garden's profit.

Try it this week

Write the garden as a linear program in GLOP and solve it (290). Then add one pupil-hour and, separately, one bed of land, and check that the profit rises by 5 and by 25. Which limit would you pay for, and up to what price?

Worksheet

Linear programming and the simplex method

Optimization · Level 4 · Lesson 1

Name _____ Class _____ Date _____

1 **In the cheapest school lunch (40.97 coins), energy and iron are met exactly, while protein comes to 23 g against a need of 20 g. What is protein's shadow price?**

◯ 3: the spare grams ◯ Zero: the limit has room to spare ◯ 41: the cost of lunch

2 **Find the shadow price of pupils' hours yourself by solving twice, then compare it with the constraint's dual_value(). Complete the program so it prints how much one extra hour adds.**

```
def best(hours):
    s = pywraplp.Solver.CreateSolver("GLOP")
    x = s.NumVar(0, 10, "tomato")
    y = s.NumVar(0, 10, "beans")
    s.Add(x + y <= 8)
    s.Add(2 * x + y <= 14)
    s.Add(x + 3 * y <= _____)
    s._____(30 * x + 40 * y)
    s.Solve()
    return s.Objective().Value()
print(best(_____) - best(18))
```

Word bank: 18 19 Maximize 17 hours Minimize

3 **The simplex method checks every corner of the feasible region before it stops.**

◯ True ◯ False

4 **Solve the diet problem with GLOP: six foods (price, kcal, protein g, iron mg per portion), at most 2 portions of each. Add the three needs as limits and the cost as the goal, then run it.**

```
from ortools.linear_solver import pywraplp
food = {"rice": (15, 260, 5, 0.4), "beans": (20, 230, 15, 4),
        "egg": (12, 75, 6, 0.9), "milk": (18, 150, 8, 0.1),
        "greens": (10, 40, 5, 6), "nuts": (8, 170, 7, 0.6)}
need = [700, 20, 5]   # kcal, g protein, mg iron
s = pywraplp.Solver.CreateSolver("GLOP")
x = {f: s.NumVar(0, 2, f) for f in food}
# the three needs as limits, then the cost as the goal
s.Solve()
print(round(s.Objective().Value(), 2))
```

Write your program:

5 **Match each term to its meaning.**

Feasible region	• • • •	Met exactly at the optimum
Binding constraint	• • • •	Crosses the inside of the region, in polynomial time
Dual value	• • • •	Every plan that keeps all constraints: a convex set
Interior-point method	• • • •	The gain from one more unit of a constraint

✎ **Think of a decision you know (a budget, a timetable, a farm plot) that could be an LP. Which limit would you expect to have the highest shadow price, and why?**

Check your answers

Linear programming and the simplex method

Optimization · Level 4 · Lesson 1

1 **Zero: the limit has room to spare**

Complementary slackness: a limit with slack has a dual value of zero.

2

```
def best(hours):
    s = pywraplp.Solver.CreateSolver("GLOP")
    x = s.NumVar(0, 10, "tomato")
    y = s.NumVar(0, 10, "beans")
    s.Add(x + y <= 8)
    s.Add(2 * x + y <= 14)
    s.Add(x + 3 * y <= hours)
    s.Maximize(30 * x + 40 * y)
    s.Solve()
    return s.Objective().Value()
print(best(19) - best(18))
```

The hours limit uses the argument, the goal is a profit to maximise, and best(19) − best(18) is the gain from one more hour: 295 − 290 = 5.0.

3 **False**

It stops at the first corner with no better neighbour.

4

```
from ortools.linear_solver import pywraplp
food = {"rice": (15, 260, 5, 0.4), "beans": (20, 230, 15, 4),
        "egg": (12, 75, 6, 0.9), "milk": (18, 150, 8, 0.1),
        "greens": (10, 40, 5, 6), "nuts": (8, 170, 7, 0.6)}
need = [700, 20, 5]   # kcal, g protein, mg iron
s = pywraplp.Solver.CreateSolver("GLOP")
x = {f: s.NumVar(0, 2, f) for f in food}
for k in range(3):
    s.Add(sum(food[f][k + 1] * x[f] for f in food) >= need[k])
s.Minimize(sum(food[f][0] * x[f] for f in food))
s.Solve()
print(round(s.Objective().Value(), 2))
```

Three ≥ constraints and a minimised cost give 40.97 coins: about 1.3 cups of rice, 0.55 cups of greens and 2 handfuls of groundnuts.

5

Feasible region — Every plan that keeps all constraints: a convex set
Binding constraint — Met exactly at the optimum
Dual value — The gain from one more unit of a constraint
Interior-point method — Crosses the inside of the region, in polynomial time

The feasible region is where the answer can be; binding constraints shape it at the optimum; dual values price those constraints; interior-point methods are the polynomial-time alternative to the simplex walk.